

High Efficiency and High Power Density Weinberg Converter Reducing Conduction Loss and Output Current Ripple for Space Applications

Dong-Kwan Kim
Division of Future Vehicle
KAIST
Daejeon, Republic of Korea
pkodower@kaist.ac.kr

Yeonho Jeong
College of Engineering and Applied
Science, Department of Mechanical
Engineering
University of Colorado Denver
Denver, CO
yeonho.jeong@ucdenver.edu

Jae-Il Baek
Department of Electrical Engineering
Andlinger Center for Energy and the
Environment
Princeton University
Princeton, NJ
jaeil.baek@princeton.edu

Jeong-Eon Park
Satellite Bus Development Division
Korea Aerospace Research Institute
Daejeon, Republic of Korea
eepark@kari.re.kr

Cheol-Woo Lim
Satellite Technology Research center
KAIST
Daejeon, Republic of Korea
sspsso@kaist.ac.kr

Gun-Woo Moon
School of Electrical Engineering
KAIST
Daejeon, Republic of Korea
gwmoon@kaist.ac.kr

Abstract— Weinberg converter is widely used to battery discharge regulator of satellite system because it guarantees soft-switching of all devices and has low output current ripple. However, due to leakage inductance of the coupled inductor and transformer, the voltage stress of powering diode and the output current spike are increased. To overcome these problems, weinberg converter with auxiliary switch and capacitor is presented in this paper. The proposed converter significantly reduces the voltage stress of powering diode. In additions, the output current spike is decreased by adopting auxiliary circuit. Thus, the conduction loss of the powering diode and output current ripple are reduced. As a results the proposed converter achieves high efficiency and high power density with the reduced output filter. The verification is performed with 400 W/50 V prototype.

Keywords—Weinberg converter, high efficiency converter, battery discharge regulator.

I. INTRODUCTION

Battery discharger regulator (BDR) is the most important power supply in the sequential shunt series regulator (S4R) system used in spacecraft application because it mainly regulates the bus voltage (V_{BUS}) which suffer entire payload [1]-[3]. As shown in fig. 1, S4R system consists of lots of modularized parts. In S4R system, series shunt regulator (S3R) and S4R supply V_{BUS} and the battery respectively. In this case, it is not possible to regulate V_{BUS} tightly through the combination of S3R and S4R. Thus, additional power supply, BDR, is used in S4R system. Because V_{BUS} should be tightly regulated by BDR, it is required to have very high reliability on dynamic response. Thus, some requirements are demanded for BDR such as step-up with low overshoot on output current and fast transient response. To satisfy them, BDR should be designed with the wide bandwidth and the high phase margin. However, lots of step-up converters based on boost converter have low transient characteristics due to the right half plane zero. Therefore, a weinberg converter, shown in fig. 2, has been widely adopted in BDR due to several advantages [8]-[11]. First, it has no right half plane zero, so that the higher phase margin with the same

bandwidth can be achievable. Second, because the operating frequency of a powering inductor (L_I) is twice larger than the switching frequency (f_s), the converter can have wider bandwidth. Therefore, the weinberg converter is very attractive considering the harsh requirement on the dynamic characteristics.

In spacecraft, reducing the weight and volume of satellite system is demanded for low energy consumption to launch. Thus, it is issued in BDR to have high efficiency to minimize a bulky heat-sink [4]-[6]. Another issue to take into account is minimizing the bus filter size. In the satellite system, the metalized polyester capacitors, MET, are basically used despite of its low power density [7]. The reason why MET is selected that the self-healing characteristics guarantees high reliable to shunt failure problem and has long lifespan. As a result, the efficient way to conceal a defect about power density is minimizing output current ripple and increasing the efficiency of BDR.

However, the weinberg converter has some drawbacks considering parasitic components. For examples, an oscillation is occurred among a leakage inductance of L_I (L_{lkgI}), that of L_M (L_{lkgM}) and a junction capacitor of one output diode D_I (C_{DI}) increases the voltage stress of D_I , thus the large conduction loss is dissipated on D_I because a diode with high voltage rating has large forward bias voltage drop. In addition, a huge output current spike presents due to L_{lkgI} and L_{lkgM} when main switches (Q_1 and Q_2) are turned off, so that the output filter size is enlarged.

Some researchers have introduced several methods to minimize the side effect [7]-[15]. [7] presented non-dissipative snubber to reduce voltage ringing of components. In this case, the suggested methods require an additional RC snubber due to regulation problems, so that the snubbing loss is still remained. Active clamping of voltage ringing components were proposed [12]-[15]. However, the high side active switches degrade the reliability of converter and the voltage stress of D_I is not clamped with the presented structures. On the other hands, methods reducing output current ripple have not treated yet.

In this paper, a new converter based on Weinberg converter to improve efficiency and power density is

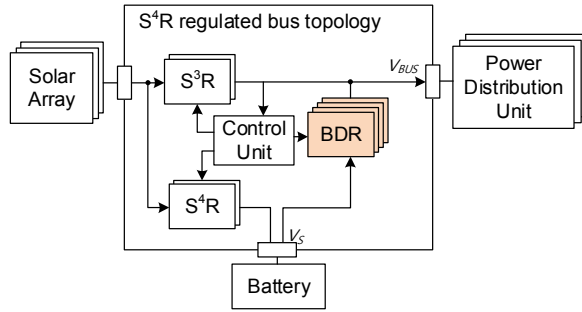


Fig. 1. S⁴R regulated bus topology.

presented. The proposed converter has several advantages as follows:

- 1) Voltage stress of D_1 is clamped to V_S . Thus, the conduction loss of D_1 is reduced by implementing the components having low forward voltage drop.
- 2) Voltage ringing over entire switching device are eliminated. Very low DM noise is occurred in a view of circuit.
- 3) A large amount of output current spike is reduced. The output filter size is decreased.

Due to abovementioned advantages, the proposed converter has high efficiency and power density comparing with the conventional weinberg converter. To verify the improvement of the proposed converter characteristics and analysis of the proposed converter are presented in the section II and III. In addition, test results with 400W/50V prototype are shown in the section IV.

II. CHARACTERISTICS OF PROPOSED CONVERTER

A. Description of Proposed Converter

Fig. 3 shows the circuit diagram of the proposed converter. In the proposed converter, an auxiliary capacitor (C_A) and an auxiliary switch (Q_A) are added to the conventional weinberg converter. In the proposed converter, both Q_1 and Q_2 operate identically with the main switches of the conventional weinberg converter and the auxiliary circuit doesn't affect to main powering operation of the weinberg converter, thus the basic characteristics such as a voltage conversion ratio and a transfer function of the proposed converter are similar with the weinberg converter. Thus, the proposed converter inherits the good dynamic characteristics of the weinberg converter.

On the other hands, C_A is regulated to V_{BUS} minus input voltage (V_S) by Q_A , thus C_A clamps the D_1 's voltage to V_S . In addition, Q_A conducts shortly after Q_1 and Q_2 are turned off. In this case, while Q_A turns on, the voltage applied to the $L_{lkgm} + L_{lk1}$ increases from $V_{BUS} - V_S$ to V_{BUS} . Therefore, the current spike caused by $L_{lkgm} + L_{lk1}$ reduces steeply. Therefore, the amount of output current ripple can be reduced. Moreover, the zero voltage switching (ZVS) operation of Q_A minimizes the additional switching loss. Finally, the C_A requires very small capacitance, so that degrades of the power density caused by the extra components are small. As a result, the efficiency and power density of proposed converter is enlarged comparing with the conventional weinberg converter.

B. Operational Principle of Proposed Converter

Fig. 4 shows the equivalent mode waveforms of the proposed converter. In the mode analysis, there are several assumptions. First, only few parasitic components, L_{lkg1} ,

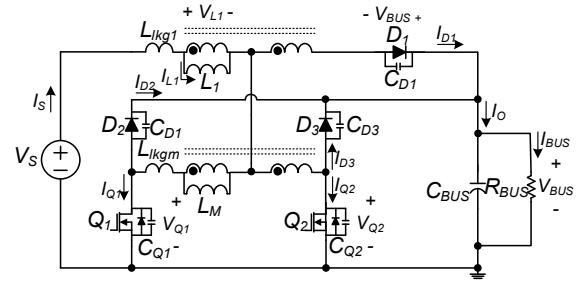


Fig. 2. Circuit diagram of weinberg converter.

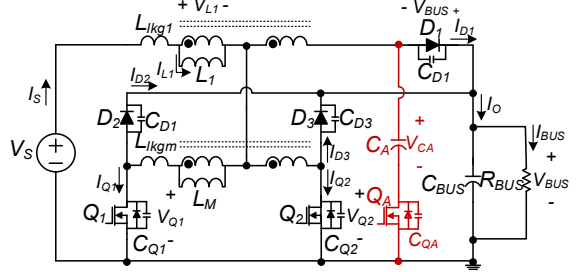


Fig. 3. Circuit diagram of proposed converter.

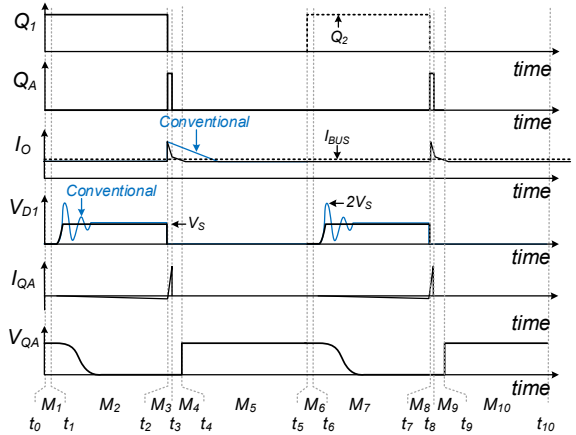


Fig. 4. Equivalent Mode waveforms of proposed converter

L_{lkgM} , output capacitors of switches (C_{Q1} , C_{Q2} , and C_{QA}), and junction capacitors of diodes (C_{D1} , C_{D2} , and C_{D3}) are concerned. Second, it is assumed that V_S , V_{BUS} , V_{CA} , and I_{L1} are constant. Third, Output current (I_O) represents $I_{D1} + I_{D2} + I_{D3}$ and bus current (I_{BUS}) means actual load current of converter. Fourth, V_{CA} is well regulated to $V_{BUS} - V_S$. Finally, Mode1~5 is identical with Mode 6~10.

Mode 1 ($M_1, t_0 \sim t_1$): Commutation of D_1 and D_2 .

After Q_1 is turned on, M_1 begins. I_{BUS} flows to Q_1 and D_2 . Due to I_{D1} is conducted, V_S is applied to $L_{lkg1} + L_{lkgM}$. Thus I_{Q1} increases and I_{D1} decreases slowly. Therefore, ZCS turn-on of Q_1 and ZCS turn-off D_1 are achieved. When I_{in} reaches $2I_{BUS}$, D_1 is cut-off and M_1 is ended.

Mode 2 ($M_2, t_1 \sim t_2$): L_1 builds up.

After $I_{in} = 2I_{BUS}$, M_1 starts. $V_S - 0.5V_{BUS}$ is applied to L_1 . Because C_A , C_{BUS} , R_{BUS} , and the body diode of Q_A operate as RCD snubbing circuit, V_{D1} clamped to V_S whereas V_{D1} of the conventional converter is ringing from zero to $2V_S$. In this case the snubbed energy of proposed converter is transferred to the output. As a result, there is no snubbing loss. Through the end of M_2 , I_{QA} , I_{D1} , I_{BUS} , and I_O are identical. M_2 is finished when Q_1 is turned -off.

Mode 3 ($M_3, t_2 \sim t_3$): Q_A conducts.

When Q_1 turns off and Q_A turns on, M_3 is started. In this case, duty of Q_A regulates V_{CA} . Due to snubbing operation during M_2 , V_{QA} is zero. Thus, Q_A turns on with ZVS. Because D_2 have conducted and D_3 is turned on immediately after Q_1 is cut-off, D_1 is also turned on. Therefore, I_O spikes from I_{BUS} to $2I_{BUS}$. While $V_{BUS}-V_S$ is applied to $L_{lkg1}+L_{lkgm}$ in the conventional converter, $2V_{BUS}-V_S$ is generated to $L_{lkg1}+L_{lkgm}$ in the proposed converter. Thus the current spike reduces largely especially in the high V_S conditions. Due to applied voltage to $L_{lkg1}+L_{lkgm}$, I_{D1} and I_{D3} increase while I_{D2} decreases slowly. As a results ZCS turn-on of D_1 , D_3 and ZCS turn-off of D_2 are achieved. M_3 is ended when Q_A turns off.

Mode 4 (M_4 , $t_3 \sim t_4$): Commutation of $D_1 \sim D_3$.

When Q_A turns off, M_4 is started. Because $D_1 \sim D_3$ are conducted, same with the conventional converter, $V_{BUS}-V_S$ is applied to $L_{lkg1}+L_{lkgm}$. Most of current spike is eliminated in M_3 . M_4 is ended when $I_O = I_{BUS}$.

Mode 5 (M_5 , $t_4 \sim t_5$): Powering of L_1 .

After eliminating of current spike D_2 is cut-off. However, D_3 is conducted continuously due to I_{LM} . Thus, V_{Q2} is clamped to V_S . $0.5(V_S - V_{BUS})$ is applied to L_1 . Most of powering current is conducted through D_1 .

III. ANALYSIS OF PROPOSED CONVERTER

A. Voltage Conversion Ratio

By adapting the voltage second balance to L_1 , we can obtain the voltage conversion ratio of the proposed converter (M_P) and that of the Weinberg converter (M_W) as follow:

$$M_P = \frac{V_{BUS}}{V_S} = \frac{1+2D}{1+2D_A}, \quad (1)$$

where D_A indicates Q_A 's turn-on duty ratio except that either D_2 or D_2 is conducted.

$$M_W = \frac{V_{BUS}}{V_S} = 1+2D. \quad (2)$$

Comparing M_P and M_W , the proposed converter can has higher voltage conversion ratio than the conventional one. However, in this paper D_A is restricted to 0 due to several reasons. The D_A term is non-linear, so that it can makes the right half plane zero when D_A varies too much. In additions, the D_A can be positive after all output diodes ($D_1 \sim D_3$) are turned off, thus, the output current ripple can be increased largely. Therefore, in this paper, by limiting D_A to be zero, the proposed converter has same dynamic characteristics with the conventional weinberg converter.

B. Duty Ratio Design of Q_A

Fig. 5 shows the conducting current on Q_A (I_{QA}) related with the turn-on duty ratio of Q_A (D_{QA}). Assuming I_2 is smaller than I_O , D_A maintained as zero. Only considering this case, we can obtain the maximum of D_{QA} as follows:

$$D_{QA} \leq \sqrt{\frac{8f_s^2 \sqrt{C_{OSS,QA}} (L_{lkg1} + L_{lkgm}) D}{M_P}}. \quad (3)$$

C. Stress Analysis

Fig. 6(a) and fig. 6(b) show the equivalent circuit model during D_1 turn-off period in the weinberg converter and that in the proposed converter respectively. As shown in fig. 6(a), the maximum voltage stress of D_1 increases to $2V_S$ in the conventional weinberg converter due to the parasitic components. On the other hands, in the proposed converter, fig. 6(b) represents that when V_{D1} reaches to V_S , there is the

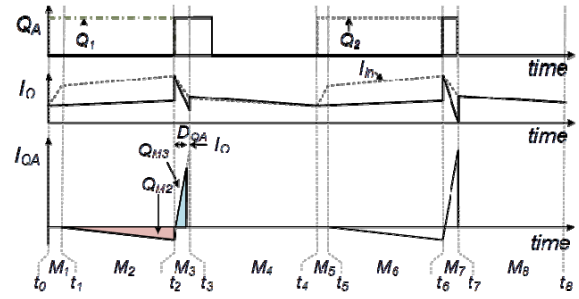


Fig. 5. Equivalent current waveform of proposed converter.

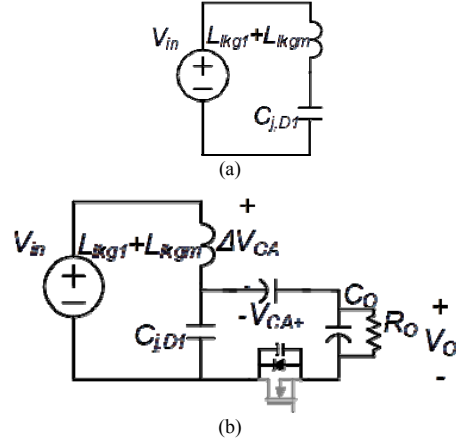


Fig. 6. Equivalent circuit during D_1 turn-off period.

(a). in the Weinberg converter. (b). in the proposed converter.

clamping current path through Q_A 's body diode. Therefore, the maximum voltage stress of D_1 is always clamped to V_S . As a results, the voltage stress of D_1 is reduced by half in the proposed converter comparing with the conventional weinberg converter.

$$Q_{SW} = \frac{I_O^2 (L_{lkg1} + L_{lkgm})}{2(V_{BUS} - V_S)}. \quad (4)$$

$$Q_{SP} = \frac{I_O^2 (L_{lkg1} + L_{lkgm})}{V_O}. \quad (5)$$

(4) and (5) are the total current spike charge of the conventional weinberg converter (Q_{SW}) and that of the proposed converter (Q_{SP}), which can be shown in fig. 6. Because voltage applied to $L_{lkgm}+L_{lkg1}$ is increased in the proposed converter, Q_{SW} is much smaller than Q_{SP} . As a result, the output filter size can be decreased largely.

IV. EXPERIMENTAL RESULTS

To verify the improvement of the proposed converter, experiments are accomplished with 27.5V-47V input and 400W/50V output conditions. In addition, the switching devices is selected considering 50% margin of maximum voltage stress. Moreover, the output capacitor (C_{BUS}) is designed considering voltage regulation conditions: <1% in steady-state operation and <5% in dynamic response.

Fig. 7(a) and fig. 7(b) show that the steady-state waveforms of the weinberg converter and the proposed converter respectively. In this case, V_S is 47V. The output current ripple of the weinberg converter is measured as 0.216A while that of the proposed converter is captured as 0.095A. Thus, the output filter, the bus capacitor, is reduced by 55% including the auxiliary circuit. In additions, the maximum voltage stress of D_1 reduced by 45%. Therefore,

TABLE I. USED COMPONENT LIST.

	Weinberg	Proposed
Q_1, Q_2	IPP045N10N3 (100V/4.5mΩ/137A)	
Q_A	IPB042N10N3 (100V/42 mΩ/137A)	
D_1	VS20CTQ150 (150V/0.85V _F)	VS16CTQ080 (80V/0.58V _F)
D_2, D_3	VS16CTQ080 (80V/0.58V _F)	
C_O	MKT373*5 (100V/2.2μF)	MKT373*2 (100V/2.2μF)
C_A	-	MKT373*1 (100V/200nF)



Fig. 7. Waveforms under 47V and 400W.

(a) Weinberg converter. (b) Proposed converter.

as shown in fig. 8, the entire efficiency is increased in the nominal input voltage (36V) of the proposed converter by adapting a diode having low forward voltage drop to D_1 .

V. CONCLUSION

This paper presents the improved weinberg converter for high efficiency and high power density. In the proposed converter, the auxiliary switch and capacitor reduces the voltage stress of D_1 largely. Thus the conduction loss of D_1 is decreased. In addition, by switching the auxiliary switch properly, the output current spike of the weinberg converter is reduced. Therefore, some output filter capacitor can be eliminated. The test results show that 55% of output filter is reduced and the full load efficiency is increased by 0.4% in the 400W/50V prototype. As a result, the proposed converter has higher efficiency and power density than the weinberg converter, i.e., the proposed converter is suitable for BDR.

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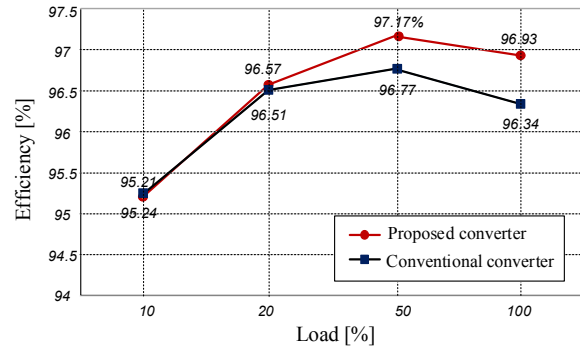


Fig. 7. Measured efficiency at 36V.

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